COVER LETTER

Raghuveer Thirukovalluru Internship in NLP/AI

From my childhood, I have been fascinated by intelligent systems and their ability to ease one's daily life. For instance, I remember how a thermostat water heater saved my parents the daily effort to carefully turn the temperature knob using screwdriver, while standing on a stool. Over time, I came across multiple such intelligent systems and have developed a strong penchant to build them. This precisely was the driving motive during my fresher and sophomore years of undergraduate studies at IIT Kanpur, where I worked on multiple projects that involved embedded systems, android development and web applications[1] to create such systems. Although I truly enjoyed building these systems, I realized that the real 'intelligence' was much greater than this. This led me to venture into the field of **Artificial Intelligence (AI)** that has catered to a huge leap in use and perception of technology and has had an impact on multiple fronts – economic, social, environmental, and more. Intrigued by the plethora of solutions offered by AI, and more specifically **Natural Language Processing (NLP)**, I undertook multiple courses and projects along these lines. With time, I have developed a very strong interest, and to some extent, a good expertise in these fields. My aspiration to understand this area in more detail and build more intelligent systems to advance state of arts has been the major motivation to my application for the internship.

My first introduction to AI was through a research project with **Dr. Nishchal Verma**, IIT Kanpur where I worked on building a framework for Condition Based Monitoring of machines. The framework is centrally based upon a novel graphical indices based feature selection technique to select best features, which are invariant to machines' changing signatures. We developed a data acquisition technique to capture these variations and proposed five visual graphical indices to quantitatively rank each feature with respect to its ability to discriminate faulty machine states from healthy ones and finally select the best set of features. Given the linearity of the quartile-quartile plots of the features, I used multivariate KL divergence equation for normal distributions to formulate a strong mathematical justification and proved the low generalization error of the features selected using our technique. In addition to this, I proved the low Vapnik-Chervonenkis(VC) dimension of the Majority Vote Classifier upon which our framework is based. I conducted rigorous experiments of tuning individual modules of feature ranking and dataset rejection to improve the performance of the system and successfully outperformed the existing arts on two health monitoring datasets. This year long exploration was reported as a **journal publication**, **IEEE Transactions on Reliability, 2017**. The entire journey strongly instilled in me, the essence of conducting research starting from literature survey to formulating mathematical justifications, empirically testing the model and finally writing a paper. This work involved long hours of onerous data collection from machines that helped me develop smart work skills and more importantly improve patience at a personal level. I strongly believe that my learnings from this very first research project would contribute to a great level in the future projects to come.

Given deep learning works well in bringing out non-linear relations between complementary features, I led the work upon using deep learning with FFT energy bins and WPT coefficients of machine's acoustic, vibration signals for machine health monitoring. To improve network stability towards noisy inputs, I initialized the deep network with weights of denoising stacked autoencoder trained using the FFT, WPT features. I further proposed a generic class median based fine-tuning technique to enhance separability between classes for multi class problems. The system achieved superior performance as compared to conventional techniques and was presented as a **conference publication** in the **IEEE International Conference on Prognostics and Health Management, (ICPHM), Canada, 2016**. Further, my eagerness to try out deep learning in other verticals led me to bioinformatics where my team from IIT Kanpur worked with **Dr. Yan Cui**, University of Tennessee on transcriptome based prostrate tumor identification. We proposed deep networks with similar loss functions and accomplished a 200 fold reduction in feature size of the gene-expression data as well as managed to achieve better performance. This work resulted in an **abstract publication** at the **15th Annual UT-KBRIN Bioinformatics, BMC Bioinformatics, USA, 2016**.

My first step towards understanding the basics of NLP was during my summer internship at Xerox Research Center India, Bangalore. I got the opportunity to work with **Mr. Ragunathan Mariappan** on building a real time summarization system for customer care conversations. I analyzed the same as a graph optimization problem and proposed a novel solution of constructing a dynamic tree where each node is a set of essential phrases extracted from a turn of the conversation. Each turn is a single exchange between the two parties in a conversation. I used handcrafted syntactic POS patterns to extract essential phrases from each turn. The resultant phrases and turns were encoded using LSTM and were later used with standard text entailment techniques to establish relations between nodes and build the tree of turns that dynamically changed as the chat progressed with time. Finally, I used the tree-knapsack algorithm alongside sub-modular utility functions for nodes to extract a sub-tree corresponding to the extractive summary of the conversation in real time. We were able to produce significant

[1] https://raghavlite.github.io/software.pdf

results on a third party dataset and also converted this work into a patent application. Moreover, we envision writing a paper on this work and hoping to publish it in a top tier conference in the near future. Finally, impressed with this work, I was offered a Pre-Placement Offer for the position of 'Budding Scientist' at XRCI along with 7 others in batch of 40 interns.

After my internship, I took an academic course in NLP under **Dr. Amitabha Mukerjee**, IIT Kanpur, where I was introduced to the basic & advanced concepts. I also worked on a course project titled 'Diachronic word sense change Identification' where we had to come up with an algorithm that could detect change in the sense of word used over a period of time. We adopted the work in 'Aligned Word Vector Spaces and Document Vectors' to train the embedding sets on different historic time ranges and proposed variance sense maps – heat maps of sense-time correlation to capture words with changed sense over time. In the last century, we found more than 50 words that had changed sense with time in two different languages - English, Hindi. During the same semester, I worked on another project in text analytics with **Dr. Harish Karnick**, IIT Kanpur. The project aimed at building a Chat-Bot that could comprehend consumer requests from the field of travel, food and generate appropriate responses for them. I designed a FrameNet inspired architecture consisting of a set of latent and functional variables attached to each state. Further, I introduced WordNet based semantic word pickers to comprehend the query by filling the variables associated to the current state; and generate responses using the corresponding knowledge before the bot transitions to the next state. Working on these projects not only solidified my concepts in the field of NLP but also motivated me to continue pursuing it after graduation.

After my graduation in July 2016, I joined XRCI as a Budding Scientist in the 'Text and Graph Analytics' group. I led the work on extracting trending topics from a stream of short text messages with **Dr. Shourya Roy**, for an Indian unicorn 'Helpshift'. I introduced a sliding window approach where each window of messages was analyzed using a three level system to mine the trending topics within it. Following prior arts, I used a semantic association based grouping technique to come up with number of topics in each window. With this as an input, I built a graphical model inspired from LDA using weighted multi-grams of phrases along with their syntactic features to find the document-topic distributions, addressing the sparsity constraints of short text with conventional topic modelling. The phrases were extracted using a custom phrase builder utilizing the semantic word groupings and n grams from the data. Finally, I used the topic-word distributions from adjacent windows to create topic mappings and came up with the trends.

As mentioned earlier, I am also inclined towards creating technical products particularly addressing public concerns. I have been an active participant in multiple coffee-room discussions and startup festivals in IIT Kanpur and in Bangalore. These explorations into understanding consumer needs, analyzing proposals and creating solutions helped me develop a clear and modular way of thinking, speaking and writing code. Along these lines, I have had strong experience in developing multiple android applications for NGOs, startups and my institute festivals[2]. Besides this, I **stood runners' up for best software award** in my final semester project on 'Local Transport Automation System'[3] amongst the graduating batch of 2016. This experience together with my coursework in software architecture, and my regular interactions with the software team at XRCI have helped me gain strong insight in developing scalable software with adaptable architecture to meet any future demand. Apart from this, I also enjoy teaching and have been a mentor for two groups in successfully driving their electronics club projects during my sophomore year at IIT Kanpur. I have also been an active member of the Deep Learning reading group at XRCI that helped my colleagues familiarize with its concepts.

Having closely worked with industry and academia on multiple research projects, I have developed a deep interest in the field of AI and NLP. My interactions with academia, fellow researchers at XRCI and co-workers from startups have further solidified this interest and fueled an aspiration of pursuing research as my career.

The major purpose of my application for an internship is to gain a rigorous background in more advanced techniques employed in AI, more specifically NLP problems. I wish to use it as an opportunity to explore multiple areas before pursuing PhD in a problem of interest. I would cherish the opportunity to work with the best researchers on different problems in fields of NLP and AI.

To conclude, I sincerely believe my strong research experience, academic background and diverse set of skills make me an ideal candidate for an internship. My inquisitive nature and alacrity to explore multiple fields, I believe could contribute to combine ideas from diverse fields to advance state of arts in science and technology. My penchant for creating products can help frame interesting research problems to work in collaboration with industry. On a different note, my chess (rated 1250 blitz, chess.com) and table tennis skills would be a good addition to the sports community at the university. On account of all the above aspects, I strongly believe that I can contribute meaningfully during the internship.

[2] https://play.google.com/store/apps/developer?id=Dazer&hl=en

[3] https://raghavlite.github.io/ltas.pdf